

Integer Programming Approaches for Real-Time Traffic Management

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ABSTRACT

Integer Linear Programming (**ILP**) is a powerful mathematical optimization technique widely used for solving complex decision-making problems across various industries. This project focuses on the application of **ILP** to real-world aviation problems, demonstrating its utility in optimizing airline operations and logistics. Two illustrative examples are explored: **the shortest path problem**, which seeks to determine the most time-efficient route between two cities, and **the minimum cost flow problem**, which optimizes the transportation of goods between source and destination nodes under constraints such as capacity and cost. Additionally, the project delves into the aviation industry in Egypt, specifically addressing strategic, tactical, and operational decisions for **Air Cairo**, a domestic airline. By applying **ILP** methods, the project aims to enhance Air Cairo's management practices, design an optimal domestic flight network, and allocate resources effectively. The results underscore the versatility and efficiency of ILP in addressing challenges within the aviation sector.

1. Introduction

In the highly competitive and resource-intensive aviation industry, efficient decision-making is critical to ensuring profitability and operational excellence. **Integer Linear Programming (ILP)** provides a structured framework for addressing a wide range of optimization problems, from routing and scheduling to resource allocation and logistics. By formulating

these problems as mathematical models, **ILP** enables decision-makers to find optimal solutions while adhering to practical constraints such as costs, time, and capacity.

This project explores the application of **ILP** to two fundamental problems in aviation logistics and network design. **The first** is the shortest path problem, which determines the quickest route between a source and a destination in a network of cities connected by flights. **The second** is the minimum cost flow problem, which involves transporting goods from

source nodes to destination nodes cost-effectively, considering supply, demand, and capacity constraints. Both problems are modeled using **ILP** and solved to highlight the potential for improving operational efficiency.

The project also examines the aviation industry in Egypt, with a case study on **Air Cairo**. The analysis investigates three levels of airline management decisions—strategic, tactical, and operational—and applies **ILP** methodologies to address specific challenges. The case study includes designing a domestic flight network connecting six major airports in Egypt, selecting hubs, and allocating aircraft resources to flight legs. By integrating **ILP** into its decision-making processes, **Air Cairo** can optimize its network design and improve its competitive position in the domestic aviation market.

Through these examples, this project demonstrates the practical applications of **ILP** in solving real-world problems in the aviation industry, providing insights into how mathematical optimization can drive efficiency and innovation.

2. Methodology

2.1. Shortest Path Problem

This problem was solved using a standard ILP solver, and the shortest route was identified between selected cities. The nodes represent the cities.

2.1.1. Parameters:

d_{ij} : Distance or travel time between city i and city j
 x_{ij} : Binary variable; 1 if the path between i and j is selected, 0 otherwise.

2.1.2. Objective Function:

Minimize the total distance:

$$\text{Minimize } Z = \sum_{i=1}^n \sum_{j=1}^n d_{ij} \cdot x_{ij}$$

2.1.3. Constraints:

- Flow Conservation:

- At the source, one outgoing path must be selected:

$$\sum_{j=1}^n x_{ij} = 1 \quad \forall i = \text{source}$$

- At the destination, one incoming path must be selected:

$$\sum_{i=1}^n x_{ij} = 1 \quad \forall j = \text{destination}$$

- For intermediate nodes, the incoming and outgoing paths must be equal:

$$\sum_{i=1}^n x_{ij} = \sum_{j=1}^n x_{ij} \quad \forall i, j \neq \text{source, destination}$$

- Binary Decision Variables:

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

2.2. Minimum Cost Flow Problem

This problem was solved to determine the optimal flow of goods in a network, minimizing costs while adhering to capacity constraints.

2.2.1. Parameters:

c_{ij} : Cost of transporting one unit of goods from node i to node j
 f_{ij} : Flow of goods between node i and node j
 s_i : Supply or demand at node i

2.2.2. Objective Function:

Minimize the total transportation cost:

$$\text{Minimize } Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} \cdot f_{ij}$$

2.2.3. Constraints:

- Flow Conservation: The total inflow and outflow must balance at each node:

$$\sum_{j=1}^n f_{ij} - \sum_{j=1}^n f_{ji} = s_i \quad \forall i$$

- Capacity Constraint: The flow on each edge must not exceed its capacity:

$$0 \leq f_{ij} \leq \text{Capacity}_{ij} \quad \forall i, j$$

2.3. Hub Location Problem

This problem was solved to optimize the selection of hub airports and allocation of non-hub airports to minimize total transportation costs for Air Cairo's domestic network.

2.3.1. Parameters:

C_j : Fixed cost of selecting j as a hub
 d_{ij} : Distance between airport i and airport j
 a_{ij} : Binary parameter; 1 if airport j covers airport i , otherwise 0
Cost per km (k): Transportation cost per kilometer.

2.3.2. Decision Variables:

$x_j \in \{0, 1\}$: Binary variable; 1 if an airport j is selected as a hub, 0 otherwise.

$y_{ij} \in \{0, 1\}$: Binary variable; 1 if an airport i is assigned to a hub j , 0 otherwise.

2.3.3. Objective Function:

Minimize the total transportation cost:

$$\text{Minimize } Z = \sum_{j=1}^n C_j \cdot x_j + \sum_{i=1}^n \sum_{j=1}^n d_{ij} \cdot k \cdot y_{ij}$$

2.3.4. Constraints:

- Hub Selection Constraint: Exactly three hubs must be selected:

$$\sum_{j=1}^n x_j = 3$$

- Assignment Constraint: Each airport must be assigned to exactly one hub:

$$\sum_{j=1}^n y_{ij} = 1 \quad \forall i$$

- Linking Constraint: An airport can only be assigned to a hub if that hub is selected:

$$y_{ij} \leq x_{ij} \quad \forall i, j$$

- Coverage Constraint: Assignments are restricted to airports within the coverage radius:

$$y_{ij} \leq a_{ij} \quad \forall i, j$$

- Binary Decision Variables:

$$x_j, y_{ij} \in \{0, 1\} \quad \forall i, j$$

3. Applications

3.1. Analysis for Shortest Path Calculation in a Directed

Graph

This problem involves finding the shortest path between nodes in a directed graph. The graph consists of five nodes labeled from A (source node) to E (destination node), each assigned a specific distance, with directed edges connecting them. The connections between the nodes and their corresponding distances are as follows :

- A to B with a distance of 4
- A to C with a distance of 2
- B to C with a distance of 1
- B to D with a distance of 5
- C to D with a distance of 8

- C to E with a distance of 10
- D to E with a distance of 2

The task is to determine the shortest route from node A to node E and compute the total distance of this path

We will solve this in the following steps:

Step 1: Construct Network Sketch

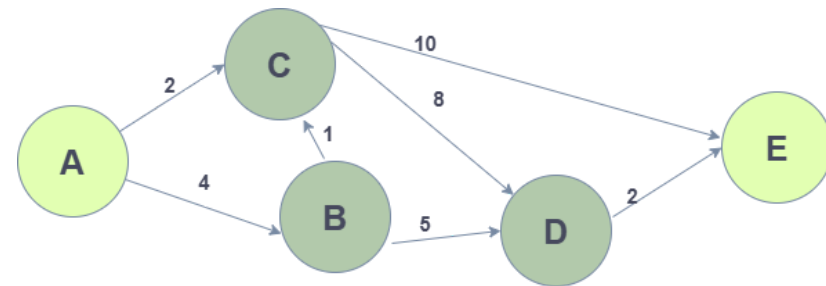


Fig.1 Network Flowchart Sketch

Step 2: Write Down Objective function and constraints

- Objective function

$$\text{Min } Z = 4x_{AB} + 2x_{AC} + x_{BC} + 5x_{BD} + 8x_{CD} + 10x_{CE} + 2x_{DE}$$

- Constraints

$$x_{AB} + x_{AC} = 1$$

$$x_{AB} - x_{BC} - x_{BD} = 0$$

$$-x_{CE} + x_{AC} + x_{BC} = 0$$

$$x_{BD} + x_{CD} - x_{DE} = 0$$

$$x_{CE} + x_{DE} = 1$$

Step 3: Solution using Excel

	XAB	XAC	XBC	XBD	XCE	XCD	XDE	Optimum value		
solution	1	0	0	1	0	0	1	11		
Objective function	4	2	1	5	10	8	2			
C1	1	1	0	0	0	0	0	1	=	1
C2	1	0	-1	-1	0	0	0	0	=	0
C3	0	1	1	0	-1	0	0	0	=	0
C4	0	0	0	1	0	1	-1	0	=	0
C5	0	0	0	0	1	0	1	1	=	1

Table.1

Step 4: Solution using MATLAB

```
>> Project_pl
The minimum distance from A to E is: 11
The shortest path from A to E is: ABDE
```

So the shortest path with optimum value =11

And route



Fig.2 Shortest path Sketch

3.2. Optimization Network Flow Cost in a Directed Graph

In this example, we explore the Minimum Cost Flow Problem in a directed graph to find the least expensive way to transport flow from a source node to a sink node. The network is defined by nodes A, B, C, and D with a directed graph with directed edges with specific capacities and costs associated with transporting flow. Here are the details :

- From node A to B, the capacity is 10 units with a cost of 2 per unit

- From node A to C, the capacity is 5 units with a cost of 4 per unit
- From node B to D, the capacity is 10 units with a cost of 1 per unit
- From node C to D, the capacity is 5 units with a cost of 3 per unit
- Node A is the source, and Node D is the sink

The objective is to determine the minimum cost of transporting any amount of flow from the source (A) to the sink (D) while adhering to the capacity limits of the edges.

We will solve this in the following steps:

Step 1: Construct Network Sketch

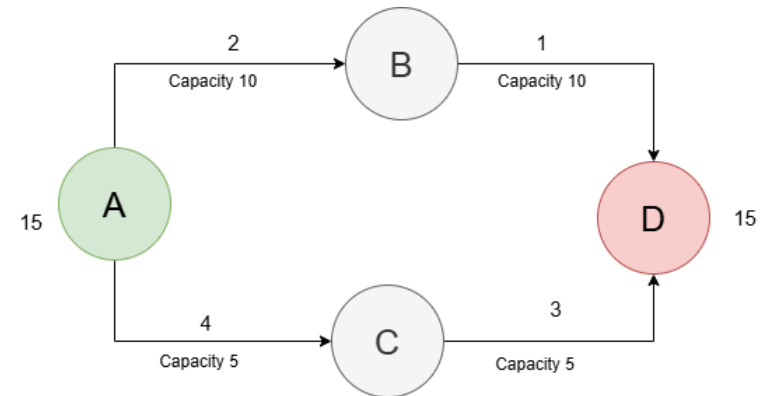


Fig.3 Network Flowchart Sketch

Step 2: Write Down Objective function and constraints

Objective function

$$\text{Min } Z = 2x_{AB} + 4x_{AC} + x_{BD} + 3x_{CD}$$

$$\text{Constraints } x_{AB} + x_{AC} = 15$$

$$x_{AB} - x_{BD} = 0$$

$$\begin{aligned}x_{AC} + x_{CD} &= 0 \\x_{BD} + x_{CD} &= 15 \\x_{AB} &\leq 10 \\x_{AC} &\leq 5 \\x_{BD} &\leq 10 \\x_{CD} &\leq 5\end{aligned}$$

Step 3: Solution using Excel

	XAB	XAC	XBD	XCD	Optimum value		
solution	10	5	10	5	65		
Objective function	2	4	1	3			
C1	1	1	0	0	15	=	15
C2	1	0	-1	0	0	=	0
C3	0	1	0	-1	0	=	0
C4	0	0	1	1	15	=	15
C5	1	0	0	0	10	<=	10
C6	0	1	0	0	5	<=	5
C7	0	0	1	0	10	<=	10
C8	0	0	0	1	5	<=	5

Table.2

Step 4: Solution using MATLAB

```
>> Project_p2
Total cost of transportation: 65
Flow distribution:
    0    10     5     0
    0     0     0    10
    0     0     0     5
    0     0     0     0
```

So the shortest path with optimum value =65
And route

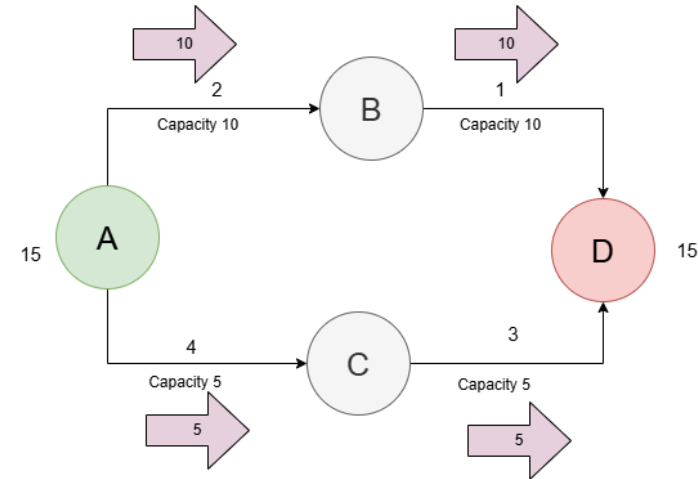


Fig.4 Network Flowchart Sketch

3.3. Optimization of Hub Selection in an Airline “Air Cairo” Domestic Network

In this example, we focus on the strategic decision of selecting optimal hubs for Air Cairo’s domestic airline network. A hub in transportation and logistics refers to a central point in a network where routes intersect, allowing for the consolidation and redistribution of passenger or freight traffic. Establishing hubs is crucial for managing the flow of traffic efficiently, reducing costs, and improving service connectivity. The network under consideration includes six Egyptian airports: Cairo International Airport (CAI), Capital International Airport (CCE), Borg Al-Arab International

Airport (HBE), Aswan International Airport (ASW), Hurghada International Airport (HRG), and Taba International Airport (TCP). The distances between these airports are specified in **Table 3**, and the objective is to select three of these airports as hubs.

Distances in KMs between these EG airports						
	CAI	CCE	HBE	ASW	HRG	TCP
CAI	0	56	219	933	467	368
CCE	56	0	275	894	431	317
HBE	219	275	0	1064	665	576
ASW	933	894	1064	0	542	1237
HRG	467	431	565	542	0	698
TCP	368	317	576	1237	698	0

Table 3 Distances in KMs between these EG airports

Constraints :

- Hub Selection Constraint: Exactly 3 airports should be chosen as hubs.
- Airport Coverage Constraint(s): Each airport should be connected to at least one of the selected hubs. We need to ensure that for each airport j , at least one of the selected hubs i must be chosen as a hub.
- Non-Negativity: The decision variables are binary (either 0 or 1).
- 1 hr. of domestic flight covers an average distance of 835 Kms $\rightarrow$ 1 hr of domestic flight costs an average of 10K USD in 2024, $\rightarrow$ 1km of domestic flight costs an average of 12 USD = 595 EGP.

We will solve this in the following steps:

Step 1: Construct Network Sketch

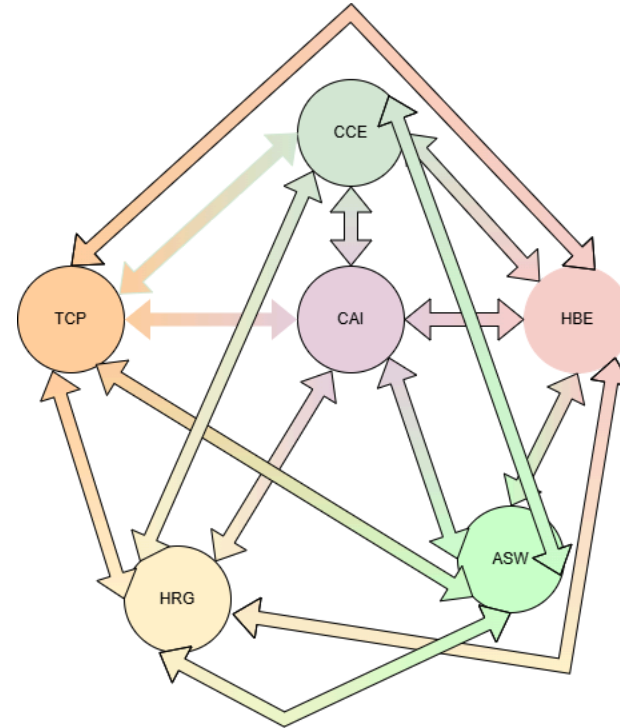


Fig.5 Network Flowchart Sketch

Step 2: Write Down Objective function and constraints

Objective Function

$$\min z = 56x_{CAI\ CCE}$$

Step 3: Solution using MATLAB

```
>> Project_p3
Selected Hubs:
    1      4      5

Minimum Cost: $7716.00
```

4. Results discussion

4.1. Analysis for Shortest Path Calculation in a Directed Graph

4.1.1. Using Built-in Function

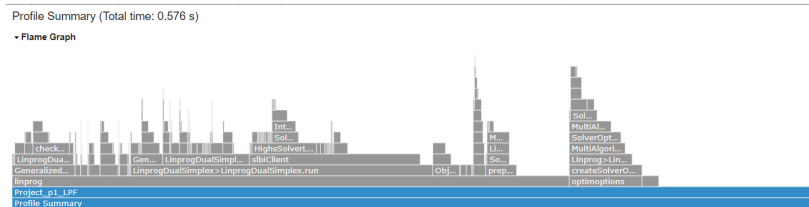


Fig.6 Profile Summary (Built-in Function)

4.1.2. Without Built-in function

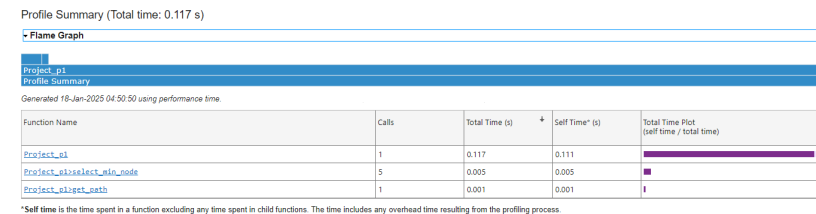


Fig.7 Profile Summary (Without Built-in Function)

After using MATLAB's Profiling Tool for our code and for MATLAB's Lindborg function, We noticed that:

The compiling time for our code is less than the compiling time for the Matlab built-in function (linprog) as our code compiling time is 0.117s but linprog compiling time is 0.576s.

This noticed a decrease in the compiling time implies that our code is more efficient and has a great performance in comparison to MATLAB's Lindborg function for these types of examples.

4.2. Optimization Network Flow Cost in a Directed Graph

4.2.1. Using Built-in Function

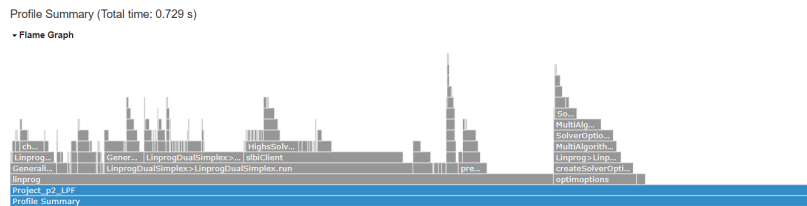


Fig.8 Profile Summary (Built-in Function)

4.2.2. Without Built-in function

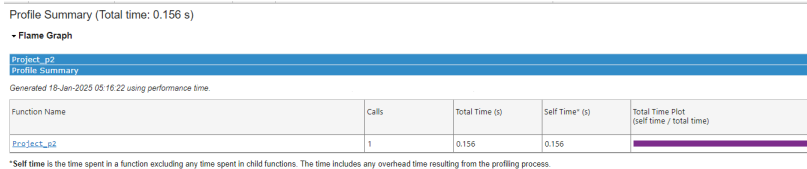


Fig.9 Profile Summary (Without Built-in Function)

After using MATLAB's Profiling Tool for our code and for MATLAB's Lindborg function, We noticed that:
The compiling time for our code is less than the compiling time for Matlab built-in function (linprog) as our code compiling time is 0.156s but linprog compiling time is 0.729s.

This noticed a decrease in the compiling time implies that our code is more efficient and has a great performance in comparison to MATLAB's Lindborg function for these types of examples.

4.3. Optimization of Hub Selection in an Airline “Air Cairo” Domestic Network

4.3.1. Using Built-in Function

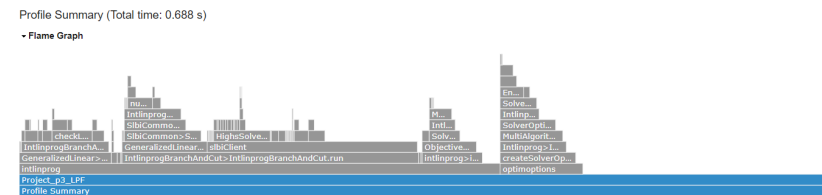


Fig.10 Profile Summary (Built-in Function)

4.3.2. Without Built-in function



Fig.11 Profile Summary (Without Built-in Function)

After using MATLAB's Profiling Tool for our code and for MATLAB's Lindborg function, We noticed that:
The compiling time for our code is less than the compiling time for Matlab built-in function (linprog) as our code compiling time is 0.181s but linprog compiling time is 0.688s.

This noticed a decrease in the compiling time implies that our code is more efficient and has a great performance in comparison to MATLAB's Lindborg function for these types of examples.

5. Conclusion

This study demonstrates the efficacy of Integer Linear Programming (ILP) in addressing complex optimization challenges within the aviation sector. By formulating and solving the shortest path problem, the minimum cost flow problem, and the hub location problem for Air Cairo's domestic network, we have showcased ILP's versatility and robustness. The application of ILP led to optimized routing, efficient resource allocation, and strategic hub selection, culminating in a significant reduction of operational costs. These findings underscore the potential of ILP as a powerful tool for enhancing decision-making processes in airline operations, ultimately contributing to improved efficiency and competitiveness in the aviation industry.

6. Appendix

The Excel sheet and all codes are attached at this [Link](#)

7. References

- [1] Barnhart, C., et al. (1998). *Flight String Models for Aircraft Fleet and Routing*.
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- [3] Taha, H. A. (2017). *Operations research: An introduction* (10th ed., Global edition). Pearson.